

Polyvinylidene Fluoride (PVDF) in Energy Storage

From Incumbent Binder to Next-Generation Functional Material

Department of Chemical Engineering

Course: CLL767 - Structures & Properties of Polymers

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Outline

- 1 Introduction & Polymer Science
- 2 Quantified Properties for Batteries
- 3 Advanced Characterization Methods
- 4 Applications: Binders & Separators
- 5 Sustainability Crisis & Alternatives
- 6 Future of PVDF & Market Analysis
- 7 Conclusions & Future Outlook

What is Polyvinylidene Fluoride (PVDF)?

A Critical Fluoropolymer

- High-performance, semi-crystalline thermoplastic fluoropolymer
- Synthesized from vinylidene fluoride (VDF): $-\text{[CH}_2\text{-CF}_2\text{]}_n$
- **Key feature:** Highly polar C-F bonds $\rightarrow$ large dipole moment
- Exceptional chemical resistance, thermal stability ($>400^\circ\text{C}$), UV resistance

Critical Role in Energy Storage

Dual Function in Lithium-Ion Batteries:

- 1 Industry-standard **polymeric binder** for electrodes
- 2 High-performance **battery separator** material

Molecular Weight Engineering

Battery grades: $M_w \sim 600,000 \text{ g/mol}^{[2]}$; Ultra-high grades: $>1,000,000 \text{ g/mol}^{[2]}$

The Key to PVDF: Crystalline Polymorphism

α -Phase (Alpha)

- Most common & thermodynamically stable
- Forms when crystallized from melt
- **Conformation:** TGTG' (Trans-Gauche)
- Dipoles cancel → **non-polar**
- Used in standard processing

β -Phase (Beta)

- Most technologically significant
- Forms via stretching or solution-casting
- **Conformation:** TTTT (all-trans)
- All C-F dipoles align → **highly polar**
- Piezoelectric & ferroelectric properties

Performance Impact

β -dominant PVDF binder: **185 mAh/g capacity & 98% retention** vs. α -phase: 71.6% retention after 50 cycles^[1]

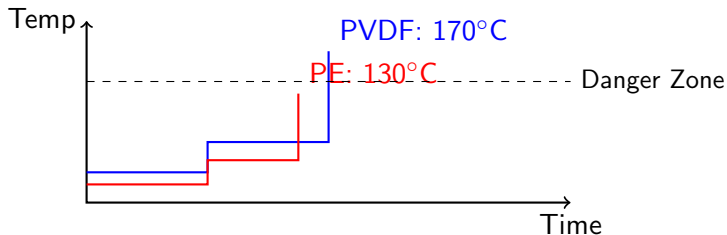
Key Thermal Properties

Glass Transition (T_g)

- **Value:** -35 to -40°C^[1]
- Far below battery operating temp
- Amorphous regions stay **flexible**
- Accommodates volume changes without cracking

Melting Temperature (T_m)

- **Value:** 165-175°C^[1,2]
- **Critical safety advantage!**
- PE/PP separators melt at ~130°C^[2]
- PVDF's higher T_m prevents catastrophic shorts during thermal runaway



The Master Property: Crystallinity

Percent Crystallinity (%)

- Typical PVDF: 50-70% crystalline^[1]
- Battery grades: engineered from 14% to 32%^[11]
- **Must balance:** strength, adhesion, electrolyte uptake

Decomposition Temperature

- **Onset:** $>400^{\circ}\text{C}$ (TGA data)^[1]
- Electrodes dried at $\sim 130^{\circ}\text{C}$ (NMP evaporation)^[9]
- **Wide processing window** ensures polymer integrity

Crystallization Kinetics (Avrami Model)

$$1 - V_c(t) = \exp(-kt^n) \text{ with } n = 1.5 - 2.4^{[1]}; T_m^0 = 182^{\circ}\text{C}^{[1]}$$

The Crystallinity Trade-Off: Data

Table: Influence of PVDF Crystallinity on Electrode Performance^[11]

Property	PVDF 1 (Low)	PVDF 2 (High)	Key Insight
% Crystallinity	14%	32%	Input variable
Adhesion	1.30 N/cm	11.42 N/cm	9× stronger!
Electrolyte Uptake	18.88%	11.3%	Lower = better
LFP Capacity Retention (500c)	136 mAh/g	146 mAh/g	Higher capacity
	64%	82%	18% better

Key Finding

High crystallinity → strong adhesion & controlled swelling → **superior long-term performance**

The Crystallinity Trade-Off: Analysis

Crystallinity → Adhesion

High-crystallinity binder has $\sim 9\times$ **stronger adhesion** due to increased intermolecular interactions between PVDF chains

Crystallinity → Swelling

Low-crystallinity binder has larger amorphous volume → only region electrolyte can penetrate → **higher swelling**

The Non-Obvious Conclusion

Excessive swelling is detrimental!

- High swelling destroys carbon distribution in electrode
- Increases internal resistance → rapid capacity fade
- **Goal:** Optimize crystallinity for adhesion + controlled swelling

Electrochemical Stability

- **The Gold Standard:** 0-5 V^[2,10]
- Stable at graphite anode (~ 0.1 V)
- Stable at high-voltage cathodes (> 4.3 V)
- One of few polymers with wide window

Dielectric Constant

- **Value:** $\epsilon = 8.4$ to 12 ^[1]
- High polarity ensures easy wetting
- Aids Li-salt dissociation
- Higher ionic conductivity

The NMP Problem

Limited Solubility: Only dissolves in polar aprotic solvents

- **NMP** (N-methyl-2-pyrrolidone)
- DMF, DMAc, DMSO

Why NMP is standard:

- High flash point ($\sim 91^{\circ}\text{C}$)
- Low vapor pressure
- Up to 99% recyclable

BUT: Classified as **SVHC** & reproductive hazard!

The NMP Challenge: A Sustainability Crisis

Why NMP is a Major Liability

- **Toxicity:** Known developmental & reproductive toxin
- **Environmental:** VOC with strict EU REACH regulations
- **Cost:** Expensive + high-energy recovery systems

Solution: Hansen Solubility Parameters (HSP)

Predictive framework for finding green solvents!

HSP decomposes cohesive energy into: dispersion (δ_D), polar (δ_P), H-bonding (δ_H)

Key Metric: Relative Energy Difference (RED)

- $RED < 1.0$: High affinity, dissolution likely
- $RED > 1.0$: Low affinity, won't dissolve

Green Solvent Alternatives: Computational Screening

HSP-Based Screening of 224 Bio-Derived Solvents

Validated PVDF model predicts solubility before costly trials!

Table: RED Values for PVDF Solvents^[5]

Solvent	RED	Assessment
NMP (Reference)	0.50	Excellent
Cyrene (Bio)	0.78	Good
γ -Valerolactone	0.80	Good
DMSO	0.88	Good

Key Insight

Cyrene & GVL are viable NMP replacements

- Thermodynamically capable of dissolving PVDF
- Bio-derived, non-toxic sources
- Identify solutions before lab testing

How Properties Are Measured

Phase & Crystallinity

FTIR: Identifies crystal phase

- Each phase has unique vibrational fingerprint
- β -phase: sharp peak at 838 cm^{-1}

XRD: Quantifies crystallinity

- Sharp peaks = crystalline regions
- Broad halo = amorphous regions
- Integrate areas to calculate %

Thermal Analysis

DSC: Measures heat flow

- T_g appears as small step
- T_m appears as large peak
- Peak area $\rightarrow$ % crystallinity

TGA: Measures mass vs. temp

- Stable plateau up to $>400^\circ\text{C}$
- Sharp drop = decomposition
- Confirms wide processing window

Slurry Rheology (Rotational Rheometer)

Critical for processability! Measures flow behavior:

- 1 **Shear-Thinning:** Viscosity $\downarrow$ at high shear (easy coating)
- 2 **Viscoelastic Gel:** $G' > G''$ at rest (prevents settling)

Mechanical Analysis

Peel Test: Quantifies adhesion

- 90° or 180° peel configuration
- PVDF: 8.7-48 N/m^[6,7,11]

DMA: Dynamic properties

- Storage modulus (G') & loss modulus (G'')
- Accurate T_g determination

Electrochemical Analysis

EIS: Impedance spectroscopy

- Measures ionic conductivity
- Identifies charge-transfer resistance

Fracture Mechanics:

- Essential Work of Fracture (EWF)
- Ductile-brittle transition analysis

Application I: The Electrode Binder

Primary Role: Cohesion & Adhesion

Despite only 2-5% of electrode weight, function is **non-negotiable**:

- **Cohesion:** Binds active material + conductive carbon together
- **Adhesion:** Adheres composite to current collector (Al/Cu foil)

PVDF's Adhesion Mechanism: Physical Only

- 1 **Van der Waals Forces:** Weak induced-dipole interactions
- 2 **Mechanical Interlocking:** Polymer chains entangle in surface roughness
- 3 **No Chemical Bonding:** Lacks -COOH/-OH groups for covalent bonds

Electrochemical Requirements

- 1 **Stability:** Inert in 0-5 V window
- 2 **Chemical Inertness:** No reaction with electrolyte
- 3 **Controlled Swelling:** Absorb electrolyte for ion transport

The Silicon Anode Problem

Why Silicon?

- Silicon offers **10× higher theoretical capacity** than graphite
- Most promising next-generation anode material

The Catch: Massive Volume Expansion

- **>300% volume expansion** during charging (lithiation)
- Graphite: only $\sim 10\%$ expansion

PVDF's Fundamental Failure

- High modulus makes it **too stiff & brittle**
- Rigid PVDF shell **cannot stretch** → **cracks**
- Leads to electrode pulverization, loss of electrical contact
- **Rapid, irreversible battery failure**

Application II: High-Performance Separator

Dual Functions

- 1 Physical Separation:** Prevents anode-cathode contact (short circuit)
- 2 Ionic Transport:** Microporous structure allows Li^+ ions to pass

Table: PVDF vs. Polyolefin Separators^[2,10]

Property	PE/PP	PVDF	Benefit
T_m	~130-165°C	~ 170°C	Higher safety
Wettability	Poor (non-polar)	Excellent	Lower resistance
Electrolyte Uptake	48-65%	92-95%	Higher capacity
Fabrication	Stretching	Electrospinning	Tunable pores

The Sustainability Crisis: NMP Problem

Toxicity & Regulation

- NMP classified as **SVHC**
- Known **reproductive toxicant**
- Heavily restricted by EU REACH
- High compliance costs

The "Aqueous Processing Revolution"

Cost Comparison:

- Water: <**\$0.02/L**
- NMP: **\$1.00-\$3.00/L** 50-150× more!

Additional NMP Costs:

- High-CAPEX solvent recovery
- Energy-intensive drying

End-of-Life Recycling Crisis

Pyrometallurgy (Burning)

Process: High-temp incineration (600-1600°C)

The Problem:

- PVDF decomposes → **HF gas**
- Hydrogen fluoride: corrosive & toxic
- Requires costly acid scrubbers
- Deactivates cathode materials

Hydrometallurgy (Leaching)

Process: Acids dissolve & separate metals

The Problem:

- PVDF's resistance = **doesn't dissolve**
- Remains as solid contaminant
- Complicates metal purification

Future Solution: SC-CO₂ Extraction

Supercritical CO₂ + DMSO co-solvent:^[3]

- Liquid density, gas diffusivity
- Dissolves PVDF, recovers materials
- No HF emission, green process

Alternative Binders: Challenging Assumptions

Common Misconception: PVDF Has Best Adhesion

Data contradicts this! Aqueous binders show **stronger adhesion**

Table: Quantitative Binder Comparison^[5,6,11,12]

Property	PVDF	PAA	CMC	Alginate
Solvent	NMP	Water	Water	Water
Adhesion (N/cm)	~1.0	~1.5	1.1-1.7	~2.0
Modulus (MPa)	600-4000	~8000	1000	119-500
T_g	-35°C	106°C	186°C	—

The Reason: Hydrogen Bonding

Aqueous binders (PAA, CMC) are rich in **-COOH** & **-OH** groups → form strong H-bonds with oxide layers. PVDF relies on weaker van der Waals forces.

The Silicon Anode Solution

PVDF Fails for Silicon

Too stiff & brittle to accommodate $>300\%$ volume expansion

PAA: The Enabling Binder

Polyacrylic Acid (PAA) is the binder of choice for Si-anodes:^[7,8]

- Unique: **high stiffness** (~ 8000 MPa) + **elasticity**
- Can withstand & accommodate extreme stresses
- Higher carboxylic acid groups $\rightarrow$ stronger Si bonds^[7,8]
- Lower swelling (7.49%) vs. SBR (9.56%)^[12]

The Aqueous Processing Challenge

pH Control is Critical:

- Alkaline slurries (pH ~ 12) corrode Al collector
- Solution: Add acid to titrate pH to 9-10

The Future: Evolving PVDF's Role

Gel-Polymer Electrolytes

PVDF-HFP copolymers:

- Absorb liquid electrolyte into polymer matrix
- Combine safety of solid with conductivity of liquid
- Higher ionic conductivity than pure solid
- Improved flexibility & processability

Composite Solid-State Electrolytes

Problem: Ceramic SSEs are brittle, poor electrode contact

- PVDF as polymer binder phase
- Disperses ceramic Li-conductors
- Improves mechanical integrity
- Enhances electrode-electrolyte contact

Biocompatibility

FDA-approved: Medical membranes, high-purity filtration

Sodium-Ion Battery Incompatibility

PVDF Fails in Sodium Systems

Fundamental electrochemical instability in Na-based systems

The Failure Mechanism

- 1 During sodiation, PVDF undergoes chemical decomposition
- 2 Produces fluoride ions (F^-)
- 3 $F^- + Na \rightarrow$ forms stable, insulating **NaF layer**
- 4 Resistive layer destroys binding ability
- 5 High resistance $\rightarrow$ rapid cell failure

Conclusion

PVDF's LIB success **does not translate to SIBs**. CMC/PAA required for Na-ion batteries.

The Solution: CMC as Superior Na-Ion Binder

Why Carboxymethyl Cellulose (CMC)?

Water-soluble biopolymer with **-COOH & -OH groups** → chemical bonding^[5,12]

Table: PVDF vs. CMC for Na-Ion Batteries^[5,12,13]

Property	PVDF	CMC	Advantage
Processing	NMP (Toxic)	Water	Green
Adhesion (N/cm)	~1.0	1.1-1.7	H-bond
Ionic Cond. (S/cm)	~10 ⁻⁵	~10 ⁻²	1000×
Swelling	11-19%	Lower	Stable
T_g	-35°C	186°C	Rigid
Na System	Fails	Stable	No NaF

Key Advantages of CMC^[5,12,13]

- **Ionic conductivity:** 1000× higher (10⁻² vs. 10⁻⁵ S/cm)
- **Chemical adhesion:** -COOH groups form H-bonds (stronger than VdW)
- **Na compatibility:** No fluoride release; no NaF formation

Market & Techno-Economic Analysis

Market Size & Growth

Driven by EV & grid storage:^[14]

- **2024:** USD 1,800.75 million
- **2032:** USD 3,605.40 million
- **CAGR:** 8.82% (2025-2032)

Key Manufacturers

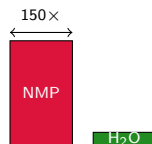
Global supply chain:^[14,15]

- Arkema (France): Kynar®
- Solvay (Belgium): Solef®
- Kureha (Japan): Battery grades
- Dongyue Group (China)
- 3M, AGC Chemicals

The Real Cost Problem

Polymer price: \$7-26/kg

- 1 **NMP Solvent:** \$1-3/L (50-150× water!)
- 2 **Solvent Recovery:** High CAPEX equipment
- 3 **Drying:** Energy-intensive (130°C)
- 4 **VOC Compliance:** Regulatory costs
- 5 **Recycling:** HF scrubbing systems



The PVDF Paradox

The Entrenched Incumbent

Unmatched Advantages:

- **5.0 V electrochemical stability** (unique)
- Thermal stability $>400^{\circ}\text{C}$
- Established supply chain
- Proven reliability at scale

The Fatal Flaws

Four Fundamental Problems:

- 1 **NMP toxicity** & environmental cost
- 2 **Weak adhesion** (physical VdW only: 8.7-11 N/m)
- 3 **Si-anode incompatibility** (brittle vs. 300% strain)
- 4 **HF recycling hazard** & non-renewable feedstock

PVDF is simultaneously essential and obsolescent

Future Research Outlook: Three Paths Forward

Path 1: "Greening" PVDF

Preserve performance, eliminate NMP

- HSP-guided solvent screening
- Cyrene, GVL ($RED < 1.0$)
- Bio-derived, non-toxic alternatives
- SC-CO₂ recycling with DMSO

Path 2: "Replacing" PVDF

Develop functional replacements

- PAA/Alginate for Si-anodes
- Chemical H-bonding (78.3 N/m)
- Self-healing networks
- Water-based processing

Path 3: "Evolving" PVDF

Leverage unique properties

- Gel-polymer electrolytes (PVDF-HFP)
- Composite solid-state electrolytes
- High-voltage cathode binder
- Functional copolymers

Strategic Direction

Bifurcated market emerging:

- PVDF: High-voltage cathodes (5V)
- Aqueous: Anodes (graphite, Si)

Key Takeaways

- 1 Crystallinity is the master property:** 32% crystallinity $\rightarrow$ 9 $\times$ adhesion, 82% vs. 64% retention
- 2 NMP is the real cost driver:** 50-150 $\times$ more than water, plus infrastructure costs
- 3 Aqueous binders have superior adhesion:** PAA $\sim$ 2 N/cm vs. PVDF $\sim$ 1 N/cm
- 4 PVDF fails for next-gen systems:** Too brittle for Si ($>300\%$ expansion), unstable in Na-ion
- 5 Future is application-specific:** Aqueous for manufacturing, PVDF for advanced electrolytes
- 6 Market is booming:** \$5.99B (2024) $\rightarrow$ \$28.75B (2033), 20.3% CAGR

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Thank You

Questions and Discussion